

Audio course on steroids during pregnancy. Episode six. Asthma, rheumatology, lupus, and maternal disease control.

Companion reading handout for simultaneous listening.

This episode is about a sentence residents must believe before they can manage pregnancy safely: treating the mother is often fetal therapy.

Steroid anxiety can make clinicians undertreat maternal disease. That is a mistake. Severe asthma, active lupus, nephritis, vasculitis, inflammatory arthritis, myositis, and other immune diseases can threaten the pregnancy through hypoxia, inflammation, preeclampsia, fetal growth restriction, preterm birth, thrombosis, hospitalization, and maternal organ injury.

So the choice is not steroid versus no risk.

The real comparison is controlled maternal disease with a suitable medication plan versus uncontrolled disease with maternal and fetal consequences.

Start with asthma.

Gabbe's respiratory disease chapter gives one of the clearest examples. In persistent asthma, inhaled corticosteroids are preferred maintenance therapy in pregnancy. The reason is not simply that inhaled drugs sound local and therefore reassuring. The reason is that airway inflammation is central to asthma, and controlling it prevents exacerbations.

Gabbe states the clinical principle plainly: it is safer for pregnant women with asthma to be treated with asthma medications than to have asthma symptoms and exacerbations.

The fetus depends on maternal oxygenation. Maternal hypoxia is not benign fetal avoidance of medication. It is fetal oxygen risk.

For a home asthma exacerbation, Gabbe gives oral prednisone forty to sixty milligrams per day, in one or two divided doses, for three to ten days.

For emergency or hospital therapy, systemic glucocorticoids are indicated when response is poor after one hour or when the patient is already taking prednisone. Methylprednisolone, prednisolone, or prednisone forty to eighty milligrams per day is used until peak flow improves, then tapered. Severe exacerbations may require one hundred twenty to one hundred eighty milligrams per day over three to four divided doses for forty eight hours before taper.

Those numbers should not be memorized as a drug list. They teach intensity. Mild disease is controlled locally. Exacerbation may need a short systemic course. Severe disease may require high-dose short-term treatment because maternal oxygenation is the immediate fetal therapy.

Now move to rheumatology.

BSR guidance identifies prednisolone as compatible with pregnancy and the preferred corticosteroid for maternal rheumatologic disease. Methylprednisolone is expected to be compatible because placental transfer is similar. ACOG immune-modulating guidance uses the same pharmacologic logic: prednisone, prednisolone, and methylprednisolone are converted by the placenta into relatively inactive metabolites.

This is the target-first model again.

For maternal disease, we generally want to treat the mother while limiting fetal exposure. That is why prednisolone-type therapy makes sense.

But steroids should not be the entire chronic plan when disease is active. Pregnancy-compatible steroid-sparing medications, such as hydroxychloroquine or azathioprine in appropriate diseases, may reduce the need for repeated high-dose rescue. A resident should ask, are we using steroids because disease flared despite a good maintenance plan, or because pregnancy-compatible maintenance therapy was unnecessarily stopped?

Now lupus.

Lupus is one of the places where steroid reasoning must be especially disciplined. Active lupus can harm the mother and fetus. But chronic high-dose glucocorticoid exposure can worsen gestational diabetes, hypertension, preeclampsia risk, infection risk, bone loss, wound healing, and adrenal suppression.

The goal is not steroid avoidance. The goal is disease control with the lowest effective steroid burden and appropriate maintenance therapy.

One important diagnostic trap is lupus nephritis flare versus preeclampsia. Both can present with hypertension and proteinuria. The difference matters most when the pregnancy is still very preterm. A lupus flare may be treatable, allowing pregnancy prolongation. Preeclampsia may require delivery depending on severity and gestational age.

Anti-dsDNA trends, complement levels, urinary sediment, hematuria, casts, fetal growth, timing, and postpartum resolution all matter. Hypocomplementemia can occur in preeclampsia too, so no single lab rescues the resident from clinical judgment.

When steroids are needed for maternal lupus flares, nonfluorinated steroids such as prednisone and methylprednisolone are preferred because placental metabolism limits fetal exposure.

Now stress-dose planning.

Gabbe's connective tissue disease chapter gives a practical threshold. Prednisone twenty milligrams per day or more for at least three weeks is a clear adrenal-suppression risk threshold. One stress-dose regimen is hydrocortisone one hundred milligrams intravenously every eight to twelve hours for two to three doses. Prednisone five milligrams per day or less has very low adrenal-suppression risk. Between five and twenty milligrams, practice varies.

The resident should not wait until labor to ask this question. If a patient is on chronic steroids, the delivery plan should mention adrenal coverage.

Now the fetal heart block exception.

Anti-Ro and anti-La disease can create a specialized fetal therapy question. Dexamethasone crosses the placenta, so it has been considered for fetal cardiac conduction disease. But the evidence is limited and controversial. Complete heart block generally should not be expected to reverse. In the PRIDE study, dexamethasone did not reverse complete heart block, and treated pregnancies had more preterm and small-for-gestational-age infants.

Some experts consider dexamethasone four milligrams per day for first-degree or second-degree block, or for hydrops, but this belongs with fetal cardiology and rheumatology expertise. It is not routine maternal lupus steroid therapy.

This distinction is essential.

Dexamethasone for fetal heart block is intentional fetal drug exposure with uncertain benefit and real potential harm. Prednisone for a maternal lupus flare is maternal disease-control therapy with lower fetal transfer. Do not blur them.

Now a bedside comparison.

A patient at ten weeks has a lupus flare. The resident should not say, steroids are dangerous in the first trimester, avoid them. That is too blunt. A better approach is, uncontrolled lupus can threaten the pregnancy. Use a lower fetal-transfer steroid when needed, at the lowest effective dose, with a plan for disease control and monitoring. Discuss the possible small oral-cleft signal honestly, but do not let a theoretical or small absolute risk lead to dangerous undertreatment.

A patient at twenty eight weeks has severe asthma, low oxygen saturation, and poor response to initial bronchodilator therapy. The resident should not hesitate because prednisone sounds scary. Maternal oxygenation is fetal therapy.

A patient with stable lupus on hydroxychloroquine and low-dose prednisone should not have maintenance therapy stopped reflexively because she is pregnant. Stopping effective pregnancy-compatible therapy may create the flare that then requires higher-dose steroids.

The resident reflex is this: for maternal disease, the fetus is protected by controlling the mother and by choosing lower fetal-transfer steroids when possible.

Do not undertreat serious maternal disease.

Do not choose fetal-transfer steroids when the fetus is not the target.

Do not rely on repeated rescue steroids because a good maintenance plan was abandoned.

And do not forget the delivery question: chronic maternal steroids may mean adrenal suppression, and adrenal suppression needs a labor plan.